

Summary

For this project, the approach taken was to implement the Page Object Model (POM) design pattern to keep page selectors, actions on these selectors, and functions organized and segmented, with user flows or more complex ones to simulate end-user behavior as accurately as possible. Using this approach allows us to maintain our tests more quickly and easily, and also allows us to scale easily since the design is divided and structured using this pattern.

The tests performed included both positive and negative tests to improve test coverage for the requested pages. Tools such as Cypress Grep were used for managing and organizing test cases, and MochaWesome Report was used to generate reports that include information about each test execution. Test Data Driven (TDD) examples were also included for scenarios where this approach was feasible and where different data was needed to run the tests.

During the development of the tests, the key challenges were researching and determining how to handle tests related to messages in pop-up windows, new tabs opened, and new windows opened after clicking on buttons that trigger these events. This was because Cypress, as such, does not handle each scenario independently. Therefore, it was necessary to simulate, for some scenarios, that the test continued on the same browser execution instance, and to validate only the messages generated in pop-up windows. Thus, it was necessary to mock these behaviors to be able to handle them correctly within Cypress.

Recommendations

As an improvement recommendation, it is suggested to integrate CI/CD with Cypress tests so that they run upon pull request and validate the integrity of new changes before merging them into the main branch, thus blocking errors in the main environment. It is also recommended to configure parallel test execution when increasing current test coverage to optimize execution time. Finally, it is recommended to use a test management tool such as Xray or Zephyr to assign IDs to test cases and thus better manage test execution reporting.